DP Biology – C1.3 (HL extension)

Inside photosynthesis

- How does photosynthesis capture and use the energy in sunlight?
- How do non-living factors affect photosynthesis?

HL statements C1.3.9–19. Textbook pp. 439–449.

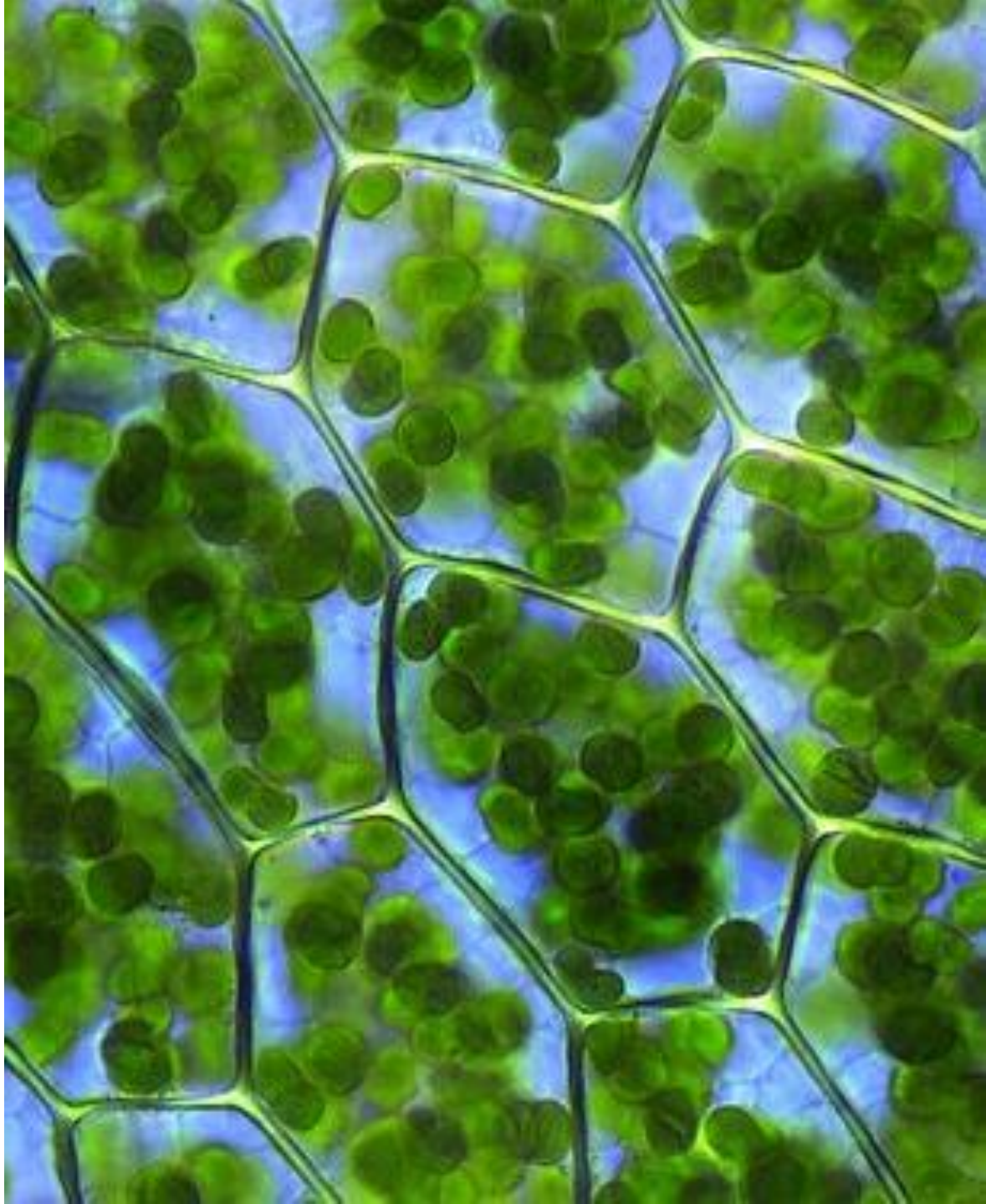
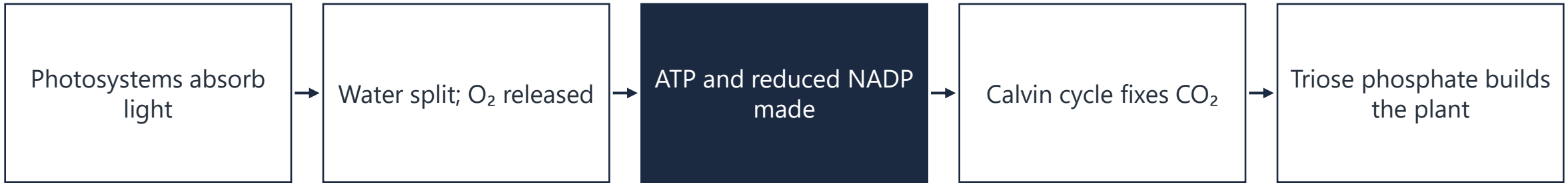


Image: chloroplasts in moss leaf cells. Kristian Peters, CC BY-SA 3.0.

Objectives for each class

Class	Statements	By the end of the class, you can...
1	C1.3.9–11	Describe photosystems, explain why pigments work as an array, and explain photolysis and its effect on early Earth
2	C1.3.12–14	Explain ATP production by chemiosmosis and the reduction of NADP, and where each happens in a thylakoid
3	C1.3.15–19	Explain the Calvin cycle (fixation, reduction, regeneration), its products, and how the two stages depend on each other

The big picture: two linked stages



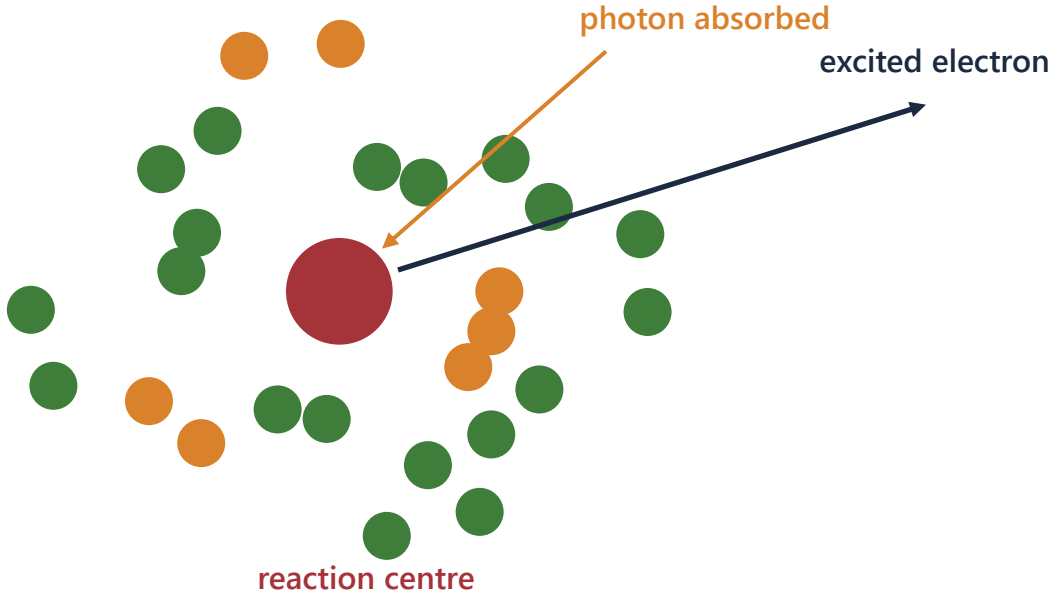
Thylakoids turn light into ATP and reduced NADP. The **stroma** uses them to fix CO_2 into sugar.

Big question

Why does the Calvin cycle, which never uses light, stop within seconds of darkness?

Photosystems

C1.3.9



A **photosystem** is an array of chlorophyll and **accessory pigments** held by proteins in the thylakoid membrane.

Energy is passed to the **reaction centre** chlorophyll, which emits an **excited electron** to an acceptor.

Two kinds

Photosystem II acts first, then photosystem I (numbered by order of discovery).

Why pigments work as an array

C1.3.10

Feature of the array	Advantage
Several kinds of pigment	Wider range of wavelengths
Hundreds of molecules	More photons captured
Precise spacing	Energy funnelled quickly to the centre
Linked electron acceptor	Excited electron captured before energy is lost

Demonstration

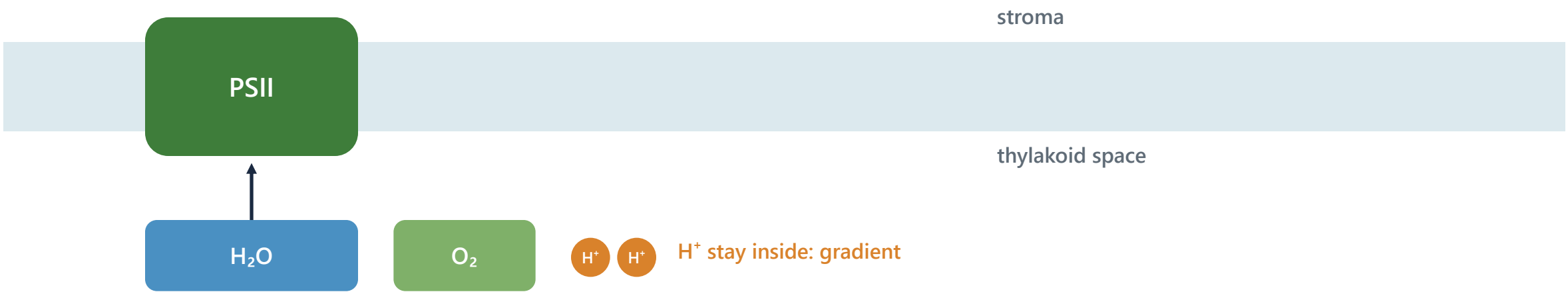
Chlorophyll in solution glows red under bright light: alone, it re-emits the energy it absorbs.

Photolysis in photosystem II

C1.3.11

$$2\ H_2O \rightarrow 4\ H^+ + 4\ \text{electrons} + O_2$$

Electrons from water replace those PSII lost.



Oxygen

A waste product; it diffuses out through the stomata. Almost all the air's oxygen came from here.

Oxygen changed the Earth

C1.3.11



Banded iron formation, Australia. G. Churchard, CC BY 2.0.

Oxygen from early photosynthesis **oxidized dissolved iron**, which settled as **banded iron**.

Then oxygen built up in the air: the **Great Oxidation Event**.

Consequences

Many anaerobes were poisoned; an ozone layer formed; aerobic respiration and complex life became possible.

A timeline of oxygen

3.5 bya — Cyanobacteria-like photosynthesis

2.4 bya — Great Oxidation Event

2.4–1.8 bya — Banded iron settles

Later — Ozone; complex life



Stromatolites, Shark Bay. CC BY-SA 2.0.

Think

Why did oxygen take hundreds of millions of years to build up in the air after photosynthesis began?

Practice: light capture

C1.3.9–11

Explain

Why an array beats single molecules.

State

The products of photolysis.

Explain

Where banded iron formations came from.

Use

reaction centre, accessory pigment, photolysis, oxidized

Check-in: class 1

C1.3.9–11

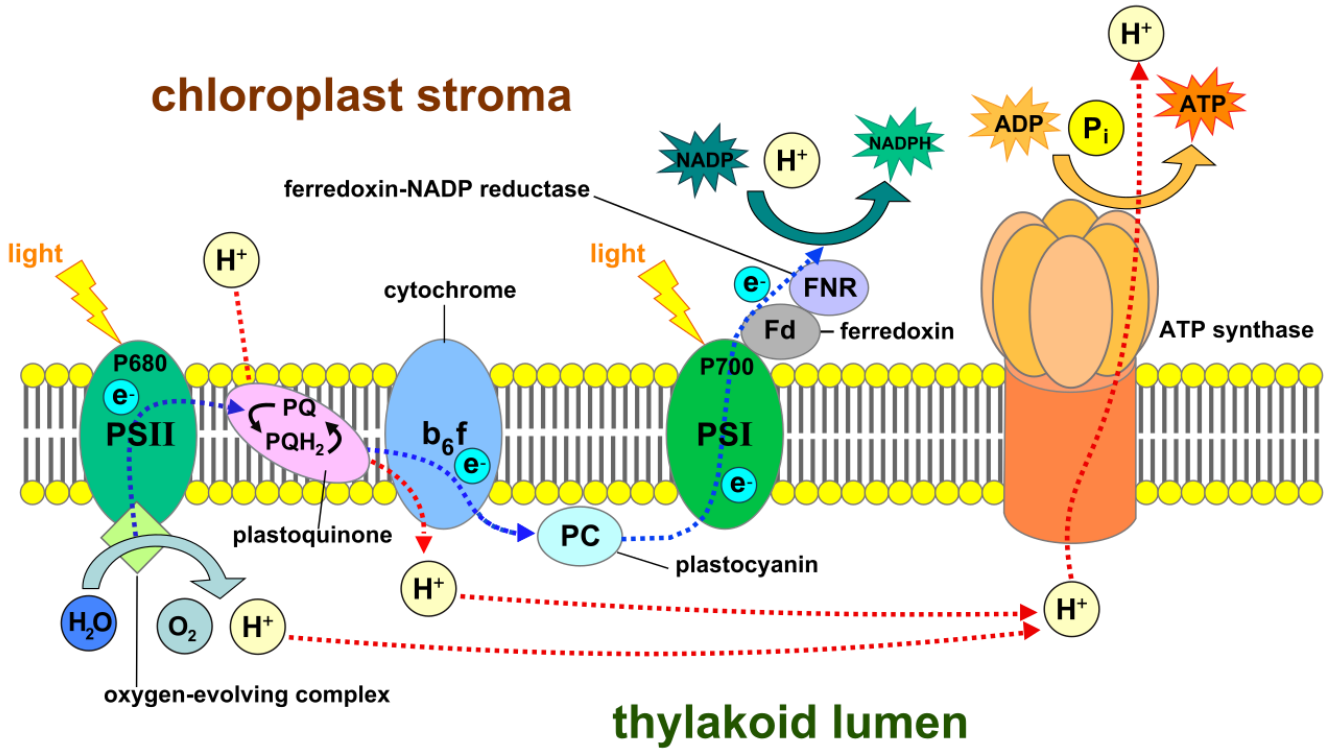
1. Define photosystem.
2. State what happens at the reaction centre.
3. Give two advantages of a pigment array.
4. Write the equation for photolysis.
5. Outline two effects of oxygen on early Earth.

How to use this

Answer without your notes, then check against slides 3–9.

ATP by chemiosmosis in thylakoids

C1.3.12



Electron carriers, proton pumping and ATP synthase. Somepics, CC BY-SA 4.0.

Electrons from PSII pass along carriers; the energy **pumps protons** into the thylakoid space.

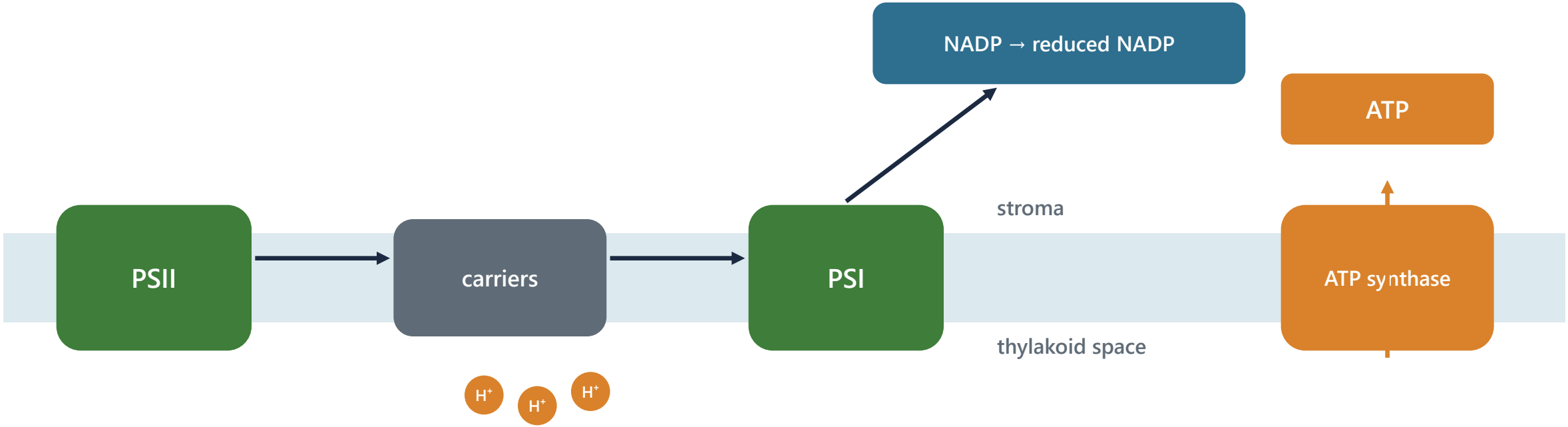
Protons flow back through **ATP synthase**: **photophosphorylation**.

Compare

Same principle as mitochondria; here protons collect in the thylakoid space.

Following the electrons

C1.3.12–13



PSII → carriers (pump H^+) → PSI → NADP. H^+ return through ATP synthase.

Key idea

Two photosystems give two energy boosts to each electron.

Photosystem I reduces NADP

C1.3.13

$$NADP + 2\ electrons + H^+ \rightarrow reduced\ NADP$$

PSI absorbs light and emits excited electrons; electrons from PSII refill it.

Two electrons and a proton reduce **NADP**. Reduced NADP carries hydrogen to the Calvin cycle.

Compare

NAD carries hydrogen away from food in respiration; NADP carries hydrogen to CO_2 in photosynthesis.

Note

NADP and reduced NADP, or $NADP^+$ and NADPH: use one pair consistently.

Cyclic and non-cyclic photophosphorylation

C1.3.12

	Non-cyclic	Cyclic
Photosystems	PSII and PSI	PSI only
Electrons from	Water	PSI, back to the carriers
Products	ATP, reduced NADP, O_2	ATP only
Why	Normal route	Extra ATP when the Calvin cycle needs it

Why both?
The Calvin cycle uses 3 ATP for every 2 reduced NADP; cyclic flow tops up the ATP.

The thylakoid as a system

C1.3.14

Process	Where
Light absorption	Photosystems in the membrane
Photolysis	PSII, facing the thylakoid space
Proton pumping	Carriers, into the thylakoid space
ATP synthesis	ATP synthase, into the stroma
NADP reduction	PSI, into the stroma

So
ATP and reduced NADP appear on the stroma side, where the Calvin cycle uses them.

Chloroplast and mitochondrion compared

C1.3.12–14

	Chloroplast	Mitochondrion
Electron source	Water	NADH and $FADH_2$
Protons pumped into	Thylakoid space	Intermembrane space
Final electron acceptor	NADP	Oxygen
ATP made by	ATP synthase	ATP synthase

Link
C1.2 HL: chemiosmosis in mitochondria.

Practice: the light-dependent reactions

C1.3.12–14

Explain

How a proton gradient forms in a thylakoid.

Distinguish

Cyclic and non-cyclic photophosphorylation.

Predict

Effect of a chemical that makes thylakoids leaky to H^+.

Use

photolysis, carriers, ATP synthase, NADP

Check-in: class 2

C1.3.12–14

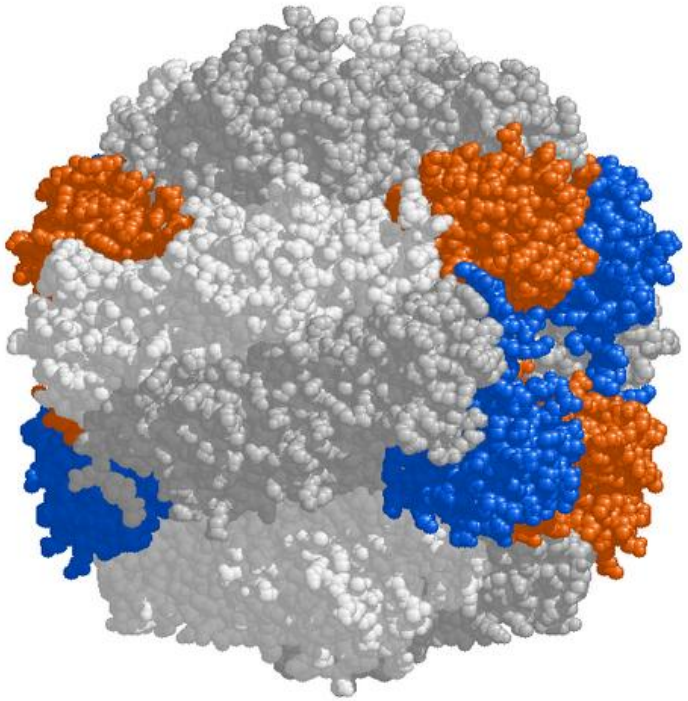
1. Explain photophosphorylation.
2. State the role of PSI.
3. Distinguish cyclic and non-cyclic flow.
4. State where ATP and reduced NADP are released.
5. Give two differences between chloroplast and mitochondrial chemiosmosis.

How to use this

Answer without your notes, then check against slides 11–17.

Carbon fixation by rubisco

C1.3.15



Rubisco. ARP, public domain.

$$\text{RuBP (5C)} + CO_2 \rightarrow \text{2 glycerate 3-phosphate (3C)}$$

Rubisco in the stroma **fixes** inorganic carbon.

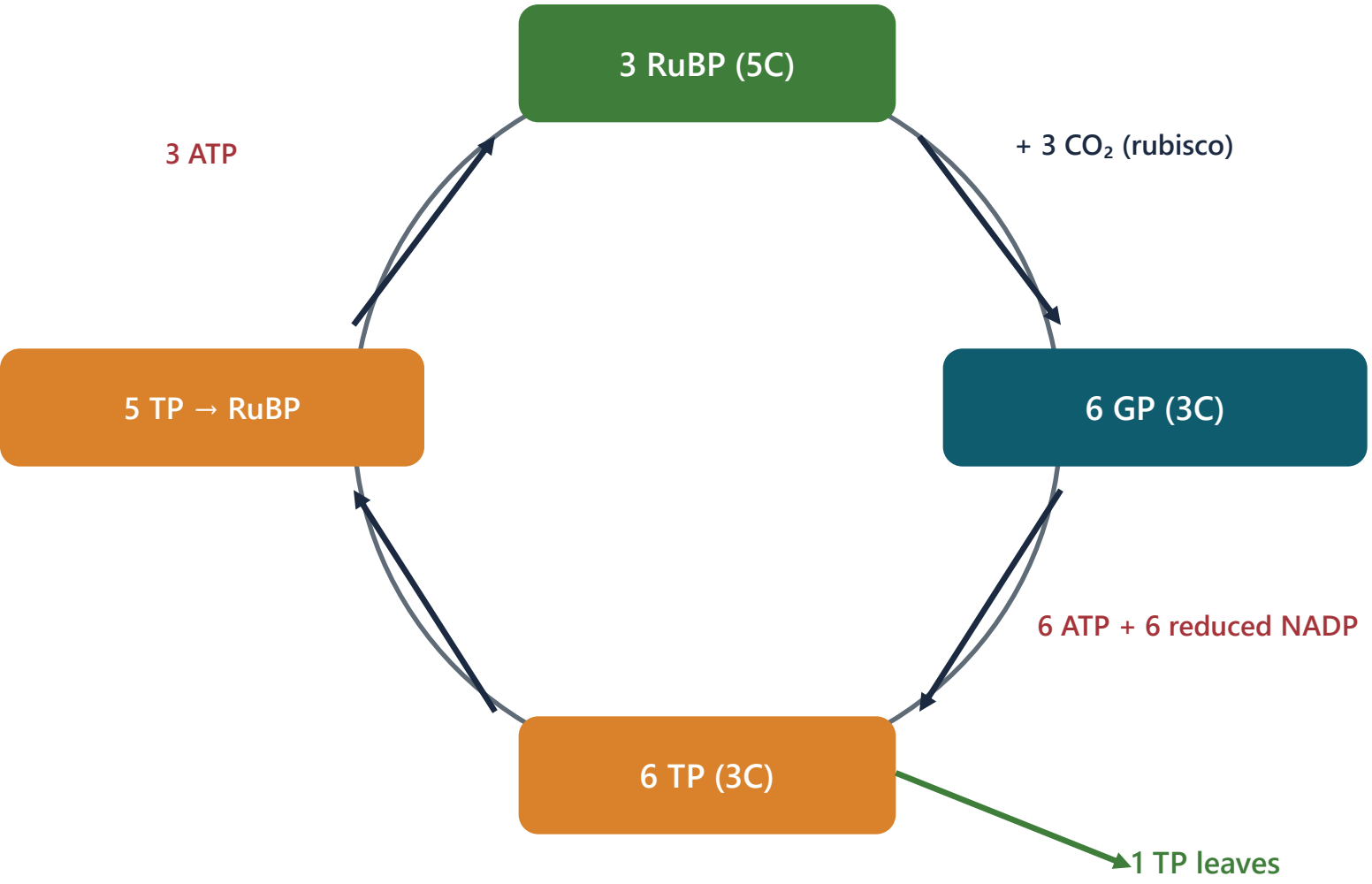
It is slow and poor at low CO_2, so chloroplasts hold huge amounts: the most abundant enzyme on Earth.

Think

Why might plants that close their stomata in hot, dry weather be limited by rubisco?

The Calvin cycle on one ring

C1.3.15–17



Fixation, **reduction**, **regeneration**.

Three turns fix 3 CO_2 and release one triose phosphate (TP).

Reduction and regeneration

C1.3.16–17

GP + reduced NADP + ATP → triose phosphate

Reduced NADP gives hydrogen; **ATP** gives energy. TP is the first sugar made.

Five TPs (15C) are rearranged into three RuBP (15C) using ATP, so the cycle continues.

Calculate

Per CO_2 fixed: how many ATP and reduced NADP? How many turns make one glucose?

Key idea

Without RuBP regeneration, carbon fixation would stop.

From triose phosphate to everything else

C1.3.18

Carbohydrates

Glucose, sucrose, starch, cellulose

Amino acids

Intermediates + nitrogen (nitrate)

Lipids

Fatty acids and glycerol

Also

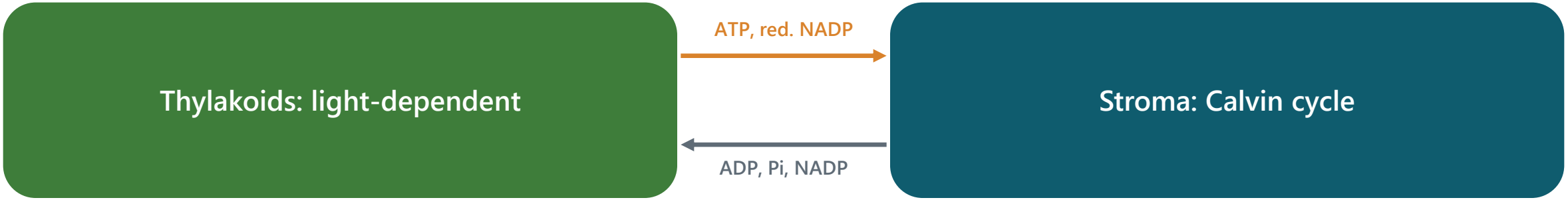
Nucleotides: sugars + nitrogen + phosphate.

Key idea

Every carbon atom in a plant was fixed by rubisco; plants also need minerals (N, P).

Two stages, one process

C1.3.19



If...	Then...
Light stops	No ATP or reduced NADP: Calvin cycle stops in seconds
CO_2 runs out	NADP not regenerated; PSI cannot pass electrons on; the chain backs up

Big question

The Calvin cycle needs the products of the light reactions.

Working with data: switching off the light

C1.3.19

Relative RuBP and GP in algae; light off at 0 s. Invented values; pattern matches classic experiments.

Time / s	RuBP	GP
-30	10	10
0 (off)	10	10
30	4	16
60	2	19
90	1	20

1. Describe the changes after the light goes off.
2. Explain why GP rises.
3. Explain why RuBP falls.
4. Predict the result of cutting CO_2 in the light.

Check-in: class 3

C1.3.15–19

1. State the substrates and product of rubisco.
2. State the roles of ATP and reduced NADP in the cycle.
3. Explain why RuBP must be regenerated.
4. Name three products made from triose phosphate.
5. Explain why the Calvin cycle stops in the dark.

How to use this

Answer without your notes, then check against slides 19–24.

Common mistakes to avoid

Not this	But this
The Calvin cycle runs at night.	It stops soon after dark.
Oxygen comes from CO_2.	From water, split by PSII.
PSI acts before PSII.	Electrons pass PSII to PSI.
Rubisco makes glucose.	Rubisco makes glycerate 3-phosphate.
All TP becomes glucose.	Five sixths regenerates RuBP.

Summary: inside photosynthesis

	Key idea
C1.3.9–10	Photosystems: pigment arrays and reaction centres
C1.3.11	Photolysis; oxygen changed the Earth
C1.3.12–14	Chemiosmosis; NADP reduced; thylakoid map
C1.3.15–17	Rubisco, reduction to TP, RuBP regenerated
C1.3.18	TP to carbohydrates, amino acids, lipids
C1.3.19	Stages depend on each other

Check that you can...

1. **Describe** photosystems and **explain** the array
2. **Explain** photolysis and early oxygen
3. **Explain** chemiosmosis, cyclic and non-cyclic
4. **Explain** the Calvin cycle stages
5. **Outline** products of triose phosphate
6. **Explain** why the two stages depend on each other

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. State the substrates and product of the reaction catalysed by rubisco.
2. Explain why photosynthesis releases oxygen.
3. Explain why the Calvin cycle stops in the dark.

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Stromatolites in Shark Bay.jpg – Happy Little Nomad – CC BY-SA 2.0
Thylakoid membrane 3.svg – Somepics – CC BY-SA 4.0
Rubisco.png – ARP at English Wikipedia – Public domain

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